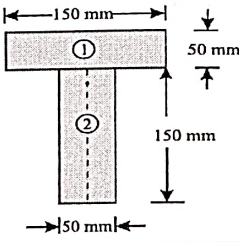


	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of BSHM Summer term Examination Academic Year: 2024-25		
Class: FY	Subject: Engineering Mechanics	Subject Code: ESC 109	
Date: 23.6.2025	Time: 10:00 to 13:00	Max. Marks: 100	

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- All questions are compulsory

Q.N.			Marks	CO	BT
Q.1	a	Explain the following i) Assumptions in analysing trusses ii) Laws of friction	10	CO4	BT1
Q.2	a	Explain the following i) Perpendicular axis theorem ii) Types of loads iii) Equation of projectile iv) Velocity v) Angle of friction	10	CO1	BT2
Q.3	a	Evaluate the moment of inertia of a T-section with flange as 150 mm × 50 mm and web as 150 mm × 50 mm about X-X and Y-Y axes through the centroid of the section. 	10	CO3	BT5
Q.4	a	The following forces act at a point: Find the magnitude and direction of the resultant force.	10	CO1	BT4

Q.5	a	A particle, starting from rest, moves in a straight line, whose equation of motion is given by: $s = t^3 - 2t^2 + 3$. Find the velocity and acceleration of the particle after 5 seconds.	10	CO5	BT4
Q.6	a	A ball is dropped from the top of a building 40 m high. At the same time, another ball is thrown upwards from the foot of the building with a velocity of 20 m/s. When and where the two stones cross each other?	10	CO5	BT4
Q.7	a	A motor car takes 15 seconds to cover 30 meters and 20 seconds to cover 42 meters. Find the uniform acceleration of the car and its velocity at the end of 18 seconds.	10	CO6	BT4
Q.8	a	Calculate forces in the members of truss.	10	CO2	BT5
Q.9	a	A body of weight 300 N is lying on a rough horizontal plane having a coefficient of friction as 0.3. Find the magnitude of the force P.	10	CO2	BT4
Q.10	a	A simply supported beam, AB of span 6 m is loaded as shown in Figure. Determine the reactions R_A and R_B of the beam.	10	CO1	BT4



SGGS Institute of Engineering and Technology, Nanded

Summer Term Examination 2024-25

F.Y. B. Tech.
Programming for Problem
Solving

All Branches
ESC 103/107

Date: 20/06/25

Time : 10:00am -1:00pm

Max. Marks : 100 Marks

- Note:
- Attempt any 10 questions from the following.
 - Assume suitable data if necessary
 - Use of non-programmable calculator is not permitted.
 - This paper contains 2 pages.

Q.N.	Questions	Marks	CO	BT
Q.1	Draw the flowchart and Write a C program to find the largest of 3 numbers.	10	CO1	BT3
Q.2	Explain any five operators used in C language.	10	CO2	BT2
Q.3	Write a calculator program in C language to do simple operations like addition, subtraction, multiplication and division. Use switch statement in your program.	10	CO2	BT3
Q.4	Write a C program to check whether a given number is palindrome or not.	10	CO4	BT3
Q.5	Differentiate the following: 1) call by value and call by reference 2) Exit controlled loop and Entry controlled loop.	10	CO4	BT4
Q.6	What is array? Write a C program for Bubble sort.	10	CO4	BT4
Q.7	Write a C program to read N numbers into an array & perform Linear search.	10	CO5	BT4
Q.8	Explain recursion in C. Write a C program in C to find Factorial of a given number using Recursion.	10	CO4	BT4
Q.9	Define string. Explain the following string functions in c with suitable program: 1) strcmp() 2) strcat()	10	CO5	BT4
Q.10	Write a program to maintain a record of a student with entities as student_name, student_no, student_marks, student_div.	10	CO5	BT3
Q.11	Explain different modes in which we can access a file using suitable programs.	10	CO5	BT3

Course Outcomes (CO)

- CO1. To formulate simple algorithms for arithmetic and logical problems.
- CO2. To translate the algorithms to programs (in C language).
- CO3. To test and execute the programs and correct syntax and logical errors.
- CO4. To implement conditional branching, iteration and recursion.
- CO5. To decompose a problem into functions and synthesize a complete program using divide and conquer approach.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Class: First

Date: / /



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Production Engineering
Summer Term Examination
Academic Year: 2024-25

Class: First Year (Gen.)	Subject: Engineering Graphics & Design	Subject Code: ESC102/106
Date: 21/06/2025	Time: 10:00 - 13:00	Max. Marks: 100

Note

- i. Attempt all questions.
- ii. Figures to right indicate full marks.

Q. 1	Solve the following questions.	M	CO	BT
a)	Draw a regular Hexagon of 60 mm side, using General method.	05	CO1	BT1
b)	Construct a regular pentagon of 50 mm diameter.	05	CO1	
c)	Draw the projection of point A situated at 30 mm away from HP and 45 mm from VP when it exists in (A) first quadrant (B) Second quadrant (C) Third quadrant (D) Forth quadrant.	05	CO1	
d)	A point R is 30 mm above the HP and 45 mm in front of VP draw it's all possible views (LHSV & RHSV).	05	CO1	
e)	What are the advantages and disadvantages of using CAD systems to create engineering drawings?	05	CO4	
f)	Explain the various basic editing commands used in Auto CAD. (Any Four).	05	CO4	
Q. 2	Solve the following questions. (Any One A or B)			
A	Draw the projections of a line AB 100 mm long inclined at 45° to VP and 30° to HP. One end of the line is 20 mm above HP and in VP. Determine apparent lengths and inclinations. Also mark the traces.	12	CO1	BT3
B	The front view and top view of a straight-line PQ measures 50 mm and 65 mm, respectively. The point P is on HP and 20 mm in front of the VP. The front view of the line is inclined at 45° to the reference line. Determine the true length of PQ and its true inclination with the reference planes. Also locate the trace.	12		BT3
Q. 3	Solve the following questions. (Any One A or B)			
A	A square lamina of 40 mm side rests on one of its sides on HP. The lamina makes 30° to HP and the side on which it rests makes 45° to VP. Draw its projections.	12	CO1	BT3
B	An equilateral triangular lamina of 50 mm sides lies with one of its edges on HP such that the surface of the lamina is inclined to HP at 60° The edge on which it rests is inclined to VP at 45° Draw its projections	12		BT3
Q. 4	Solve the following questions. (Any One A or B)			
A	A square prism 35mm sides of base and 65 mm axis length rests on HP on one of its edges of the base which is inclined to VP at 30° . Draw the projections of the prism when the axis is inclined to HP at 45° .	14	CO1	BT3
B	A cylinder 35 mm diameter and 60 mm axis, its resting on one point of a base circle on HP while its axis makes 45° with HP and T.V. of the axis 35° with VP. Draw projections.	14		



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Production Engineering
Summer Term Examination
Academic Year: 2024-25

Class: First Year (Gen.)

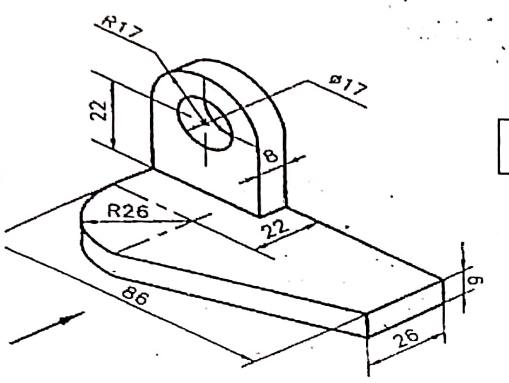
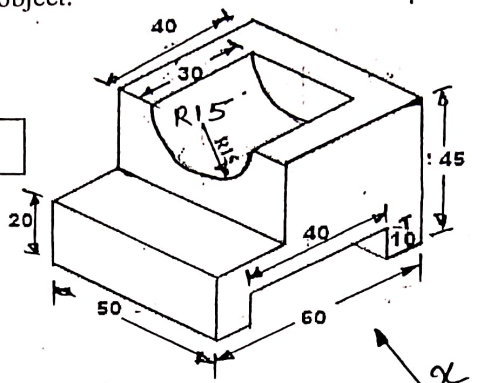
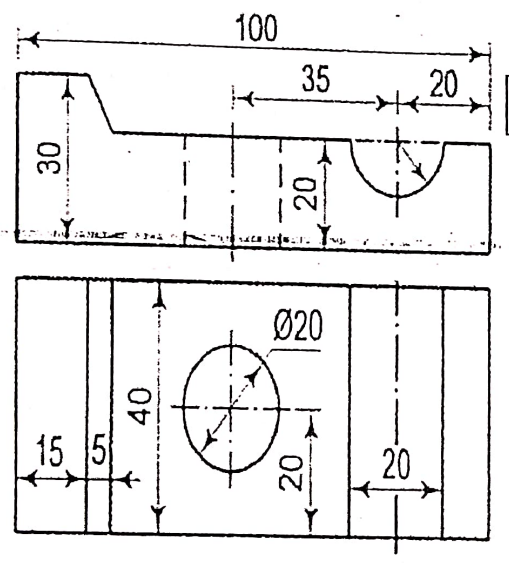
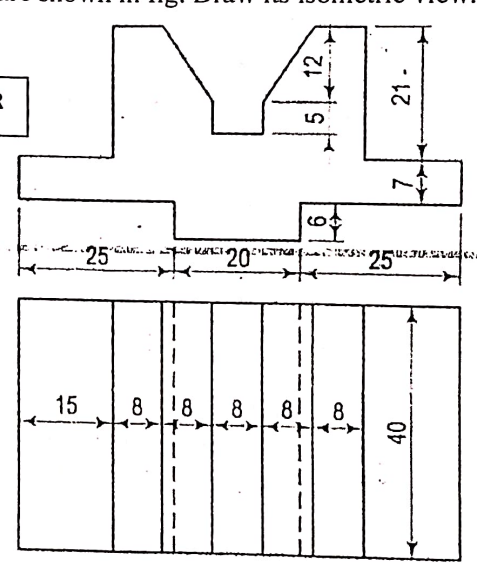
Subject: Engineering Graphics & Design

Subject Code: ESC102/108

Date: 21/06/2025

Time: 10:00 - 13:00

Max. Marks: 100

Q.5	a) Using First Angle Projection Method draw – (a) Front View (b) Top View (c) Right Hand Side View for the following object. 	b) Using First Angle Projection Method draw – (a) Front View (b) Top View (c) Left Hand Side View for the following object. 	16	CO2	BT6
Q.6	a) The Front and Top view of an object are shown in fig. Draw its isometric view. 	b) The Front and Top view of an object are shown in fig. Draw its isometric view. 	16	CO3	BT6

Course Outcomes:

CO1. Draw projections of point, line, plane and solid using the fundamental principles.

CO2. Read, interpret, and draw orthographic projections.

CO3. Read, interpret, and draw isometric projections.

CO4. Draft various Geometrical Elements used in Engineering Practice using CAD software.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

Nanded

ESC102/106
s: 100

BT6



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Summer Term Examination 2024-25

F.Y. B. Tech Div: All (A to J)

Department: BSII&M

Subject: Chemistry

BSC102

Date: 19/06/2025

Time: 10.00 am - 1.00 pm

Max. Marks: 100

- Note: i. Attempt all questions
ii. Assume suitable data if necessary.
iii. Use of a non-programmable calculator is permitted.
iv. This paper contains 2 pages.

Data: Atomic Weight: Ca = 40, H = 1, O = 16, Cl = 35.5, S = 32, Mg = 24, N = 14, Si = 28, C = 12

Q.N.	1	Questions	Marks	CO	BT
		Question 1 is Compulsory	15		
a		A sample of coal contains C = 82 %, O = 8 %, H = 6 %, S = 0.5, N = 0.2 %.. Calculate GCV & NCV of a coal sample using Dulong & Petit formula.	3	CO4	BT4
b		Classify the following impurities in to temporary, permanent and non-hardness causing impurities Ca(HCO ₃) ₂ , MgSO ₄ , CaCl ₂ , NaCl, Mg(HCO ₃) ₂ , CaSO ₄ .	3	CO2	BT2
c		Enlist any six principles of Green Chemistry.	3	CO3	BT1
d		Differentiate between Galvanising and Tinning.	3	CO5	BT2
e		Enlist the important applications of carbon nanotubes in various fields.	3	CO1	BT1
		Attempt any three	12		
Q.N	2		4	CO2	BT3
a		Define corrosion. Explain how the rate of corrosion of the following metal is influenced by atmospheric oxygen. i) Molybdenum ii) Sodium	4	CO4	BT4
b		A KMnO ₄ solution of unknown concentration is placed into a spectrophotometer. A student finds that the solution's absorbance is 0.920. The molar absorptivity of KMnO ₄ is 7.35 Lit mol ⁻¹ cm ⁻¹ and the path length of the solution container is 5.00 cm. What is the concentration of solution?	4	CO3	BT2
c		Give the important properties & uses of Fullerene	4	CO1	BT1
d		Find the acid value of an oil sample whose 6 ml required 0.02 N KOH for titration. (Density of the Oil = 0.91). State whether the oil is suitable for lubrication or not?	4		
		Attempt any three	15		
Q.N.	3		5	CO4	BT4
a		Define saponification value, give the reaction for saponification. Find the saponification value of a vegetable oil sample, whose 5 gms was saponified using excess alcoholic KOH (0.5 N). The mixture required 15 ml of 0.5 N HCl while blank titration required 45.0 ml of same HCl.	5	CO3	BT2
b		A solution of KMnO ₄ of concentration 3.5 x 10 ⁻⁵ M shows an absorbance of 1.3 at its wavelength of λ _{max} . Calculate the concentration of an unknown KMnO ₄ solution which shows an absorbance of 0.86 in the same cuvette.	5	CO2	BT1
c		Distinguish between vulcanized and non-vulcanized rubber.	5	CO1	BT4
d		Explain with a neat diagram the zeolite process of water softening including the following points. i) Principle, ii) Process, iii) softening & regeneration reactions.	5		
		Attempt any three	18		
Q.N	4		6	CO5	BT3
a		Define Lubricant & Lubrication. Explain thick film fluid mechanism of lubrication with neat labelled diagrams.	6		

b	1.50 gm of coal sample was taken for C and H estimation by combustion method. The increase in weight of tube containing anhydrous CaCl_2 and KOH bulb was found to be 1.25 gm and 3.92 gm respectively. Calculate the % of C and H.	6	CO3	BT3
c	State Pilling Bedworth rule, Explain different types of oxide layer formed during dry corrosion.	6	CO2	BT2
d	1.2 gm of CaCO_3 was dissolved in HCl and the solution was made to 1000 ml with distilled water. 20 mL of the above solution required 35 mL of EDTA solution on titration. 50 mL of the hard water sample required 30 mL of the same EDTA solution on titration. After boiling 50 mL of this water, required 12.5 mL of EDTA solution, Calculate temporary, permanent, and total hardness.	6	CO4	BT4
Q.N.	5	Attempt any Four	40	
a	What is proximate analysis? Give its Significance. 2.5 g of coal sample was exactly weighed into the silica crucible. After heating for 1 hour at 110°C the residue weighed 2.368 g, the crucible next covered with a vented lid and strongly heated for exactly seven minutes at $950 \pm 20^\circ\text{C}$. The residue weighed 1.75g. The crucible was then heated without cover at 700°C until a constant weight was obtained. The last residue was found to weigh 0.95g. Calculate the percentage results of above analysis. (% Moisture, % Volatile matter, % Ash and % F.C.).	10	CO2	BT2
b	i) 2.5 gm of blended oil was saponified using excess alcoholic KOH (0.5 N). After refluxing the mixture was titrated against 0.5 N HCl. The burette reading was found to be 24 ml. The blank titration required 40 ml of the same HCl solution. Find the saponification value of the oil. If the oil used for blending has a saponification value of 188, calculate % oil in the blend. ii) 2.5 gm of coal sample was analyzed for nitrogen content by Kjeldhal's method. The liberated ammonia required 12.7 ml of 0.5 N H_2SO_4 solution for neutralization. In a separate experiment, using a bomb calorimeter 1.5 gm of coal sample gave 0.28 gm of BaSO_4 . Calculate % N & S in the coal sample.	10	CO3	BT3
c	What are the different methods for controlling corrosion? Explain cathodic protection method in detail along with its type.	10	CO5	BT2
d	Give the preparation, properties and uses of following polymers 1) PMMA 2) Kevlar	10	CO3	BT1
e	What are the advantages & Disadvantages of Lime & Soda method? Calculate the quantity of lime (90 % Pure) and soda (95% pure) required for softening of 50000 liters of water containing the following impurities. $\text{Ca}(\text{HCO}_3)_2 = 81$ ppm, $\text{MgCl}_2 = 95$ ppm, $\text{CaSO}_4 = 68$ ppm, $\text{SiO}_2 = 50$ ppm, $\text{Mg}(\text{HCO}_3)_2 = 146$ ppm, $\text{H}_2\text{SO}_4 = 49$ ppm.	10	CO1	BT4

Course Outcomes (CO)

- CO1. Students will be able to understand various industrial processes and products.
CO2. Enabling students to know about the relevant engineering materials, their chemistry and recent technology.
CO3. Outer world will be benefited with current scientific knowledge.
CO4. Students will be able to understand various analytical methods used for material analysis.
CO5. To strengthen the student's understanding to the related core subjects in their respective discipline.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create